PHASE P · Session 3

Market Mode Detector — SHADOW ship + 5-day observation window

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POLYMARKET 5-minute BTC up/down bot
Module: **src/polymarket_bot/strategies/market_mode_detector.py**
Tests: **29 new (697 total green)**
Bot HEAD: **fd56b61** on **claude/phase-P-hard-reset**
Session 3 ship: **2026-05-17 22:41:25 UTC** · MainPID **1023320**

MMD live in SHADOW mode — annotates fires, never abstains

Phase Q decision opens Day-5: promote shadow → gate, iterate params, or shelve.

Contents

§1	Implementation summary
§2	Test inventory (29 new)
§3	Pre-deploy gate evidence
§4	VPS deploy walkthrough
§5	Day-0 verification
§6	5-day observation window + Phase Q decision matrix
§7	Rollback procedure + failure-mode contract

§1 Implementation summary

§1.1 What shipped

File	Change
src/polymarket_bot/strategies/market_mode_detector.py (NEW)	MarketModeDetector class + MmdReading dataclass + MmdBinanceKlinePoller. ~327 LO
src/polymarket_bot/strategies/spot_aware_v2_1.py (modified)	Constructor gains mmd_getter / mmd_mode / mmd_labels_to_abstain. _maybe_fire annot
scripts/run_paper.py (modified)	_build_strategy reads MMD_MODE + MMD_LABELS_TO_ABSTAIN from env at construc
pyproject.toml + uv.lock	Added ta>=0.11.0 to runtime deps.
tests/strategies/test_market_mode_detector.py (NEW)	29 tests — see §2.

§1.2 Public API (the only surface bot code touches)

MarketModeDetector:

- `MarketModeDetector(lookback_min=60, adx_strong_threshold=25.0, adx_weak_threshold=20.0, bb_compressed_bps=30.0, momentum_threshold_bps=10.0)`
- `push_bar(open_, high, low, close, volume, ts_ns) → None`
- `clear_buffer() → None`
- `bar_count() → int`
- `latest(now_ns=None) → MmdReading(label, confidence, adx, bb_width_bps, momentum_bps, sampled_at_ns, bar_count)`
- `Labels: UP / DOWN / CHOPPY / MIXED / UNKNOWN. UNKNOWN is ALWAYS pass-through.`

MmdBinanceKlinePoller:

- `MmdBinanceKlinePoller(mmd, symbol='BTCUSDT', poll_interval_s=30.0)`
- `poll_once() → int (bars added, raises on HTTP error / malformed JSON)`
- `run(stop: asyncio.Event) → coroutine (poll loop, never raises; logs warnings on failures)`

Strategy:

- `SpotAwareStrategyV2_1(..., mmd_getter=None, mmd_mode='shadow', mmd_labels_to_abstain=())`
- `set_mmd_getter(getter) → None (rebinds post-construction; used by run_paper.main())`

§2 Test inventory — 29 new

tests/strategies/test_market_mode_detector.py:

Category	Tests
Detector init	init_defaults_ok / validates_lookback_min / validates_adx_thresholds / validates_positive_bps_thresholds
Buffer	push_bar_accumulates_to_lookback / clear_buffer_empties / unknown_when_buffer_under_30
Labels	up_on_synthetic_uptrend / down_on_synthetic_downtrend / choppy_on_compressed_flat / mixed_when_neither_trend
Confidence	confidence_always_in_unit_interval / latest_returns_unknown_on_classifier_exception / no_op_stub_when_ta_missing
Poller	poll_once_happy_path / poll_once_http_error_propagates / poll_once_rejects_malformed_payload / run_polls_then_st
Strategy wire	mmd_getter_none_path_unchanged / shadow_annotates_but_never_abstains / gate_abstains_on_configured_label / g

Notable: the strategy-side tests use a duck-typed _FakeReading instead of importing MmdReading so the strategy ↔ MMD dependency stays one-way. The 'gate_treats_unknown_as_passthrough' test explicitly puts UNKNOWN in the abstain set and verifies the fire still happens — that's the design §6 HARD failure-mode contract.

§3 Pre-deploy gate evidence — all green

Gate	Evidence
pytest	667 passed (was 638 pre-Session 3; +29 new)
mypy --strict src/	Success: no issues found in 84 source files
ruff check	All checks passed!
import-linter	Contracts: 3 kept, 0 broken — production never imports paper / paper never imports live runtime, health, storag
uv lock	Resolved 101 packages; Added ta v0.11.0
pre-commit hooks	ruff + ruff-format + forbid TODO/FIXME/hack + enforce module docstrings — all PASSED on commit

§4 VPS deploy walkthrough

§4.1 Sequence (chronological, VPS UTC)

UTC	Action
22:39 UTC	git push origin claude/phase-P-hard-reset — pushed fd56b61
22:41:05	VPS git fetch + git pull --ff-only origin claude/phase-P-hard-reset — 6 files changed, 945 insertions(+), 1 deletion(-)
22:41:10	VPS .envrc backed up to .envrc.bak.phase-P-session-3-pre-mmd
22:41:11	Appended `export MMD_ENABLED=true` + `export MMD_MODE=shadow` to .envrc (other MMD_* defaults)
22:41:25	sudo systemctl restart bot-staging.service — systemd unit's `uv run` rebuilt + installed ta==0.11.0 automatically
22:41:26	MainPID=1023320 ActiveState=active SubState=running. Startup log shows mmd_mode=shadow / MMD ENABLED line /
22:41:27	bankroll_getter wired (start=$1000.00, replay=True, cap=1) — replayed to current=$790.15 (post-Session-1 drawdown pr
22:42:49	12 fires today, all pre-restart (no mmd annotation yet); next opportunity to annotate = window 1779057900 = 22:45 UTC

§4.2 Startup log signature

```
spot_aware_v2_1: kelly_mult=0.15 max_bet_abs=$500 multi_fire_cap=1
tier_relax_p_win_fallback=None tier_relax_price_floor=0.420
block_up_in_tiers=('70-85%',) hour_blocklist_utc=(22, 23, 0)
hour_blocklist_tiers=('70-85%',) mmd_mode=shadow mmd_labels_to_abstain=()
active strategy: spot_aware_v2_1
instant-settle ENABLED: gamma poll timeout=30.0s interval=5.0s
post-hoc gamma retry ENABLED: initial_delay=300.0s interval=30.0s
spot_aware_v2_1: bankroll_getter wired (start=$1000.00, replay=True, cap=1)
bankroll replay: starting=$1000.00 current=$790.15
MMD ENABLED: lookback_min=60 poll_interval_s=30.0 adx_strong=25.0
adx_weak=20.0 bb_compressed_bps=30.0 momentum_bps=10.0
warmer started; lookahead=12 windows
GET https://api.binance.com/api/v3/klines?symbol=BTCUSDT&interval;=1m&limit;=60 200 OK
```

§5 Day-0 verification

Verification protocol for the first fires post-Session-3 ship:

	Check
✓	bot-staging.service ActiveState=active / SubState=running (MainPID 1023320)
✓	Startup log emits `MMD ENABLED:` line + `spot_aware_v2_1: ... mmd_mode=shadow`
✓	First Binance kline poll succeeds (HTTP 200; 60 bars buffered)
✓	MMD task `mmd-binance-poller` running alongside warmer / runner / halt-watcher
■	First post-restart fire's rationale ends with `mmd=<label> mmd_conf=<conf>` (window ≥ 1779057900 = 22:45 UTC)
■	Dashboard decision tape surfaces the `mmd=` annotation in the rationale field
■	0 mmd-related ERROR lines in journalctl over Day-0

Items marked ■ are pending the next fire; bot fires sparingly (J.1.6 spec) so it can take 5-30 minutes between fires. The Day-0 audit script (re-runnable any time): `grep '"kind": "fired"' state/decisions/2026-05-1*.jsonl | python3 -c 'import json,sys; [print("[MMD]" if "mmd=" in (j:=json.loads(l)).get("rationale","") else "[NO]", j["window_ts"], j["side"]) for l in sys.stdin]'`

§5.1 Live bot state at Session-3 ship close

MainPID	1023320 since 2026-05-17 22:41:25 UTC
HEAD	fd56b61f4a75134980ba8ec5f69114ae3c0cff15
Branch	claude/phase-P-hard-reset
Strategy	spot_aware_v2_1 (Kelly 0.15× / replay / cap=1 / J.1.5 fold / J.1 floor / J.2 hour blocklist / H.2.12 pa
MMD config	MMD_ENABLED=true / MMD_MODE=shadow / labels_to_abstain=() / lookback=60min / poll=30s
Bankroll	starting=$1000.00 / current=$790.15 (post-Session-1 drawdown)
Today fires (pre-restart)	12 (all WITHOUT mmd annotation — old MainPID 1019607)
Today fires (post-restart)	0 (waiting for first window-fire opportunity)

§6 5-day observation window + Phase Q decision matrix

§6.1 Rules during observation

• NO env flips during the 5-day window (anti-vibe rule).

• NO src/ ships during the 5-day window.

• Operator reviews dashboard daily; flags any UNKNOWN-rate > 5% or 0 mmd annotations on fires.

• Day-5 audit runs an extended phase_p_mmd_backtest.py against the new live fires + compares to Session-2 backtest numbers.

• Halt sentinel (operator manual halt) is the ONLY allowed mid-window intervention.

§6.2 Phase Q decision matrix (Day-5)

Day-5 signal	Action
MIXED WR ≥ next-best label by ≥ 10pp on n ≥ 100 live fires	PROMOTE shadow → gate: set MMD_LABELS_TO_ABSTAIN=CHOPPY,UP,DO
MIXED WR delta 5-10pp	KEEP SHADOW; iterate on indicator parameters (try MMD_ADX_STRONG_THR
MIXED WR delta ≤ 5pp	SHELVE MMD. Reconsider strategy class entirely (Phase R pivot question).
UNKNOWN-rate > 5% (poll failure / indicator exception)	DIAGNOSE at Phase Q before any promote-to-gate. Hardening needed in poller
MMD annotation missing from any fire's rationale	BUG. Investigate why mmd_getter returned None / threw / wasn't wired. Halt and

§7 Rollback + failure-mode contract

§7.1 Rollback (any time, < 60 s)

```
ssh ops@
cd ~/polymarket-5m-bot
sed -i '/MMD_ENABLED/d; /MMD_MODE/d' .envrc
sudo systemctl restart bot-staging.service
```

When MMD_ENABLED is absent (or false), run_paper.py never creates the poller task and never binds the getter. The strategy's mmd_getter stays None and `_maybe_fire` skips the MMD code path entirely — byte-identical to J.1.6.

§7.2 Failure-mode contract (design/P.1 §6)

Failure	Behavior
`ta` library missing at init	MarketModeDetector.__init__ logs WARN and sets _ta_available=False. latest() returns UNKNO
Binance poll network/HTTP failure	Poller logs warning, retries with backoff. Buffer keeps last-good bars. After ~2 × poll_interval of s
Indicator computation exception	Caught at latest(); returns UNKNOWN. Strategy treats as pass-through.
MMD_ENABLED=true but ta missing in env	Bot still boots (MMD becomes no-op stub). startup logs WARN about ta. Operator sees this imm
MMD_MODE=gate but MMD_LABELS_TO_ABSTAIN empty	Boot validation rejects; shadow (no abstain). Bot does not fail to start.
Strategy reads UNKNOWN label	ALWAYS pass-through. Even in gate mode with UNKNOWN in abstain set. HARD requirement.
Strategy mmd_getter raises exception	Caught at fire-time; treated as UNKNOWN; fire proceeds.

End of Phase P Session 3 ship. Bot LIVE on J.1.6 + MMD shadow on claude/phase-P-hard-reset@fd56b61. Phase Q decision opens Day-5 from this ship.